

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

PROJECT SYNOPSIS

B. Tech CSE (AIML) — 7th Semester

Group No.:	50
Project Title:	GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense
Area / Domain:	Generative AI & AI Safety (Trustworthy AI / MLOps)
Academic Session:	2026-2027
Project Guide:	Dr Abhishek Kaushal

Details of Project Team:

S. No.	Enrollment No.	Name	Signature
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Abstract / Project Summary (at least 250 words):

As Generative Large Language Models (LLMs) like ChatGPT, Claude, and Llama 3 are rapidly integrated into production enterprise applications (such as medical virtual assistants, financial chatbots, and customer support engines), two critical vulnerabilities pose severe operational risks: (1) Adversarial Prompt Injections and Jailbreaking attacks that trick LLMs into revealing system secrets or breaking safety policies, and (2) Model Hallucinations where LLMs confidently generate non-factual, contradictory statements. Existing commercial guardrails (such as Meta Llama-Guard-3 or offline NLI fact-checkers like SelfCheckGPT) suffer from severe latency bottlenecks, adding 1 to 3 seconds of delay per request or evaluating responses only after full generation completes—making real-time streaming user experiences unfeasible.

To solve these fundamental limitations, GuardShield AI introduces a high-performance, dual-stage sidecar proxy microservice designed to inspect both inputs and output streaming tokens with sub-25ms latency overhead. In the Pre-Scan stage, incoming user prompts are evaluated by a fine-tuned DeBERTa-v3 context-aware security classifier to detect and block prompt injection attacks, system prompt exfiltration attempts,

and safety policy violations before they reach the target LLM (returning an immediate HTTP 403 Security Alert). In the Post-Scan stage, as the LLM streams response tokens back to the user, GuardShield continuously monitors token probability logit entropy in real time. When token uncertainty spikes past a calibrated threshold, an asynchronous windowed Natural Language Inference (NLI) Cross-Encoder evaluates sentence-level factual entailment against reference context mid-sentence, enabling early-token termination before hallucinated content is fully delivered.

GuardShield AI is engineered as a 100% software-based, zero-cost solution that runs locally on standard developer hardware using Ollama (Llama-3.1-8B) and PyTorch MPS / CUDA acceleration. The project will be evaluated on benchmark datasets (JailbreakBench, HackAPrompt, HaluEval) alongside our team's original synthetic dataset (GuardShield-Bench). The 7th-semester deliverable includes the core fine-tuned model, token entropy algorithms, and an IEEE research paper submission, while the 8th-semester extension adds a production async FastAPI streaming proxy, an autonomous LLM Red-Teaming agent, and a live Streamlit/React telemetry dashboard.

Methodology to be adopted:

The project follows a structured engineering methodology divided into 5 modular stages:

1. **Dataset Ingestion & Merging Pipeline:** Concatenate peer-reviewed open datasets—JailbreakBench (NeurIPS 2024), HackAPrompt (EMNLP 2023), and BeaverTails (NeurIPS 2023) for prompt injection defense; HaluEval (EMNLP 2023) and FEVER (NAACL 2018) for hallucination detection. Additionally, our team will construct GuardShield-Bench-v1, an original dataset of 5,000+ synthetic obfuscated attacks (Base64, Hinglish code-mixing).
2. **Model Architecture & Fine-Tuning:** Fine-tune microsoft/deberta-v3-small (86M parameters) using PyTorch and Hugging Face Trainer for multi-class pre-scan classification. Fine-tune cross-encoder/nli-deberta-v3-base for NLI factual entailment scoring.
3. **Streaming Logit Entropy Inspector:** Implement real-time Shannon Entropy computation over streaming token probability distributions: $H(x) = -\sum(p * \log(p))$. Trigger localized windowed NLI entailment checks mid-sentence only when entropy spikes past threshold H_thresh .
4. **FastAPI Sidecar Proxy Development:** Build an asynchronous HTTP / Server-Sent Events (SSE) proxy server in FastAPI intercepting request payloads and streaming responses.
5. **Telemetry Analytics Dashboard & Red-Teaming:** Build a Streamlit / React dashboard displaying live request throughput, proxy latency overhead (ms), and blocked request logs. Develop an autonomous Red-Teaming LLM agent (Qwen-2.5-14B) for automated adversarial testing.

Resource requirement (Hardware & software etc):-

- **Hardware Requirements:** 100% Software-Based (Zero Physical Hardware Purchases Required). Runs locally on standard laptops (MacBook M4 Pro with 24GB RAM / HP Envy with RTX 3050 GPU).
- **Local LLM Engine:** Ollama (Running Llama-3.1-8B and Qwen-2.5-14B locally).

- Deep Learning Libraries: PyTorch, Hugging Face Transformers, Datasets, Accelerate, ONNX Runtime (PyTorch MPS / CUDA GPU acceleration).
- Backend & Proxy Framework: Python 3.11+, FastAPI, Uvicorn (Async SSE streaming), Docker.
- Frontend & Visualizations: Streamlit / React + TailwindCSS for real-time security telemetry.
- Total Monetary Cost: ₹0 (Fully open-source stack and local execution).

Justification of the project:-

1. High Enterprise Demand: Every company deploying GenAI chatbots (in banking, healthcare, customer support) requires robust security guardrails against jailbreaks and hallucinations.
2. Fills Critical Research Gap: Solves the latency bottleneck of existing offline checkers (SelfCheckGPT, EMNLP 2023) by enabling sub-25ms mid-sentence early token-termination during live streaming.
3. 99% Parameter Reduction: Proves an 86M parameter SLM proxy matches the security accuracy of Meta's 8B Llama-Guard-3 while running locally on standard laptop hardware.
4. Publication Potential: Provides clear quantitative evaluation metrics suitable for 7th-semester IEEE/Springer student conference submissions.

PERT Chart/Schedule of project completion:-

Phase / Timeline	Tasks & Deliverables	Status / Target
Weeks 1 - 3 (Sem 7)	Literature Survey, Environment Setup, Ollama & PyTorch MPS Configuration	In Progress
Weeks 4 - 7 (Sem 7)	Dataset Merging & GuardShield-Bench-v1 Original Synthetic Dataset Creation	In Progress
Weeks 8 - 12 (Sem 7)	Fine-Tuning DeBERTa-v3 Classifier & Token Logit Entropy Algorithm Implementation	Target Sem 7
Weeks 13 - 16 (Sem 7)	Benchmarking Evaluation & 7th Semester IEEE Research Paper Submission	Target Sem 7
Weeks 17 - 30 (Sem 8)	FastAPI Async Proxy, Autonomous Red-Agent, Streamlit Dashboard & Docker Demo	Target Sem 8

References: Research papers / books / websites etc.:-

1. Manakul, P., Liusie, A., & Gales, M. J. F. (2023). SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. Proceedings of EMNLP 2023, DOI: 10.18653/v1/2023.emnlp-main.557.
2. Chao, P., et al. (2024). JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models. NeurIPS 2024, DOI: 10.52202/079017-1745.
3. Inan, H., et al. (2024). Llama Guard: LLM-based Input-Output Safeguard for Human-AI Conversations. Meta AI Research, arXiv:2312.06674.

4. Li, J., et al. (2023). HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models. EMNLP 2023, DOI: 10.18653/v1/2023.emnlp-main.397.

5. Zhang, E., et al. (2025). CAPTURE: A Context-Aware Prompt Injection Benchmark for Large Language Model Guardrails. ACL 2025, DOI: 10.18653/v1/2025.acl-main.482.

6. Lin, S., et al. (2022). TruthfulQA: Measuring How Models Mimic Human Falsehoods. Proceedings of ACL 2022, DOI: 10.18653/v1/2022.acl-long.229.

Signature(s) of project team: _____

Name & Signature of Guide:

Date: _____